

Research Paper

Enhancing energy efficiency and thermal comfort in residential buildings through stochastic model predictive control

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ABSTRACT

Model predictive control (MPC) is an advanced control strategy to minimize energy consumption while ensuring thermal comfort in residential buildings. This methodology utilizes a grey-box model, where the parameters that reflect the influence of various external and internal factors play a crucial role. The main novelty of this study is the development of a method that offline integrates parameter identification and scenario-based stochastic MPC to handle uncertainties without online adaptation. The proposed approach includes four key stages: first, the development of resistor-capacitor model to capture the thermal dynamics of the building; second, the implementation of a parameter identification model, generating multiple scenarios based on historical data; third, the proposal of a stochastic MPC framework that includes these parameter scenarios along with their associated probabilities, thereby improving building thermal performance amidst uncertainty; and finally, a performance comparison between the developed stochastic framework, deterministic MPC, and a baseline Proportional-Integral (PI) controller focusing on energy costs and thermal comfort. A sensitivity analysis further investigates the impact of electricity pricing and the trade-off between energy use and occupant comfort. The proposed approach is tested and validated using the BOPTTEST open-source testing framework, incorporating an air-to-water heat pump system coupled with floor heating. Results demonstrate that the stochastic MPC substantially improves energy management: it achieves a 50% lower discomfort penalty than deterministic MPC, reduces total cost by 24.2% compared with the PI controller over a two-week evaluation period, and cuts computation time by 70.4% through silhouette-guided k-means clustering. These findings highlight the significant potential of stochastic MPC for enhancing building energy efficiency, occupant comfort, and real-time feasibility.

1. Introduction

To achieve net-zero carbon building stock by 2050, the International Energy Agency (IEA) forecasts a need for a 50% reduction in direct building sector emissions and a 60% decrease in indirect emissions by 2030 [1]. Residential buildings offer substantial opportunities for improving energy efficiency and load flexibility [2]. One effective approach is the implementation of advanced control algorithms, particularly Model Predictive Control (MPC) [3]. This technique predicts and optimizes building performance at each time step using a mathematical model that considers future disturbances and comfort constraints [4]. Therefore, developing reliable building models is crucial for effective MPC implementation [1].

There are three primary methods for developing a building energy

model: data-driven black-box models, physics-based white-box models, and semi-physical hybrid grey-box models [5]. Among these, semi-physical models have gained significant attention in research, as they accurately represent the thermal dynamics of buildings without requiring a detailed description of heat transfer processes [1].

To develop a reliable grey-box building energy model, a three-step approach has been proposed, which includes model structure selection, parameter estimation, and model evaluation [6]. Once the model order is determined, the parameters defining the effects of various factors must be estimated using parameter identification methods [1]. However, unmeasured uncertainties and model uncertainties can significantly challenge the effectiveness of developed energy models and consequently MPC performance [7].

An effective MPC framework relies on an underlying model that approximates the actual system, often with parameters manually set by

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Nomenclature		$\dot{Q}_h$	Heat influx brought by the heating system (W)
Abbreviations		Q_{int}	Internal heat load due to occupants, lighting, and indoor equipment (W)
BOPTTEST	Building optimization building framework	R_{in}	Resistance to heat exchange between the outdoors and envelopes ($\frac{K}{W}$)
HP	Heat pump	s	Slack variable that stands for deviation from acceptable state range, silhouette coefficient
HVAC	Heating, Ventilation, and Air Conditioning	$\bar{S}$	Average silhouette coefficient
KPI	Key performance indicator	t_s	Start time of the evaluation period
MPC	Model predictive control	t_f	End time of the evaluation period
PI	Proportional-Integral controller	T_a	Outdoor temperature (K)
RBC	Rule-based controllers	T_{in}	Measured Indoor temperature (K)
RC	Resistor-Capacitor	$\tilde{T}_{in,t}$	Modeled Indoor temperature (K)
Variables and parameters		u	Control input
a	Mean Euclidean distance	x	Data point
A	Floor area	Superscripts	
b	Minimum mean Euclidean distance to samples in other clusters	i	Data point
C_{in}	Thermal capacitor of the zone and the envelopes ($\frac{J}{K}$)	j	Scenario number
C_k	Cluster k	t	Time step
H	Control horizon	Subscripts	
h	Step size	g	Ground truth
I_{solar}	Solar irradiation ($\frac{W}{m^2}$)	k	Number of clusters
J	Total number of scenarios	max	Maximum
k^*	Optimal number of clusters	min	Minimum
K_e	Coefficient of occupants' heat gain (—)	t	Time-step
K_s	Effective area of the windows (m^2)	Greek letters	
N	Optimization horizon, Number of zones	β	Decision variable that remains consistent across scenarios
N_{max}	Total number of original scenarios	μ_k	Average value of points in cluster k
$Prob$	Probability function	w	Weighting factor for trade-off between energy price and thermal discomfort
P_e	energy price		
P_{hp}	Heat pump power consumption (W)		
P_{total}	Total heat pump power consumption (W)		
Q_{hp}	Heat pump heat generation (W)		

building engineers and remaining static throughout the building's life. This approach has two main limitations: sub-optimality and non-adaptivity [8,9]. Additionally, sensor measurement uncertainties can affect control effectiveness, particularly when sensors lack precision or exhibit bias, such as when a thermostat is placed near a heat source or in direct sunlight, leading to inaccurate temperature readings [8,10].

Given the high implementation costs associated with MPC, there is an urgent need to quantify model-related and unmeasured uncertainties to improve prediction accuracy in building energy models. Ultimately, this will facilitate the development of transferable MPC structures that can effectively manage these uncertainties.

1.1. Literature review

1.1.1. Building thermal model

To improve building energy efficiency, building energy models play critical roles in various tasks, including fault detection and diagnosis, demand-side management, energy management, the integration of distributed and renewable generation resources, microgrid development, and control optimization [1].

For effective implementation of model predictive control and classification systems in buildings, a straightforward modeling method is essential that can produce interpretable parameters while accurately predicting future thermal behavior. Traditional physics-based models often become too complex due to the intricate nature of building structures and may require specialized software for simulation. Additionally, existing buildings may not align with their original blueprints, complicating the development of accurate physics-based models. In contrast, data-driven models, which use simpler structures and calibrate

parameters from actual building data, generally offer better prediction performance but lack physical interpretability [11].

A viable compromise between physics-based white-box models and data-driven black-box models is the use of grey-box thermal network models. These models, typically represented as resistor-capacitor (RC) analogues, combine fundamental physical principles of building thermodynamics with a limited number of parameters calibrated from observational data. However, they still carry significant epistemic uncertainty due to model approximations and unrecognized disturbances, alongside aleatoric uncertainty from measurement noise [1,11]. Given these uncertainties, the present study adopts a 1R1C grey-box model and focuses on quantifying the epistemic uncertainty in its parameters through historical data, rather than assuming fixed values.

1.1.2. Indoor thermal control approaches

Various control strategies have been developed for managing indoor temperature in buildings, each with its own advantages and limitations. The simplest of these, on/off Control, regulates heating and cooling systems based on deviations from a setpoint, turning them on or off as needed. While this method is intuitive and easy to implement, it fails to account for the building's thermal dynamics, leading to inefficiencies and limited user control over energy consumption. An improvement over the on/off method is weather-compensated control, which utilizes outdoor temperature sensors to adjust indoor heating according to a predefined curve. Although this approach responds to external conditions, it similarly neglects the dynamic behavior of the building, limiting its effectiveness [12].

Proportional-Integral-Derivative (PID) control is another widely adopted feedback mechanism that adjusts indoor temperatures based on

the error between the current state and a reference setpoint. This method incorporates thermal dynamics, making it more effective than the previous strategies. However, PID control requires complex parameter tuning and does not explicitly address model uncertainties, which can affect system performance [12].

Commonly used control strategies in buildings, such as on/off and PID control, are straightforward to implement and operate reliably; however, they are reactive and cannot adapt to environmental changes. To address these limitations, researchers have investigated advanced control strategies that optimize multi-objective functions for better energy management. Among these, Model Predictive Control (MPC) has emerged as the most extensively studied method due to its predictive capabilities, allowing it to adapt to changing conditions. MPC has garnered significant attention in the building industry for its effectiveness in improving indoor comfort while simultaneously reducing energy consumption [13].

MPC manages dynamic systems by predicting future states within a defined timeframe. It calculates a sequence of control actions at each sampling interval, executing only the first action while leveraging predictive data for improved performance. In HVAC systems, the MPC controller aims to minimize operating costs and maintain indoor temperatures close to the desired setpoint. By balancing these objectives, MPC enhances efficiency and comfort in system operation [4].

MPC schemes depend on precise forecasts of future system dynamics [14] and underlying model [15] to effectively identify the optimal control actions necessary for achieving desired performance outcomes. The application of MPC to HVAC systems necessitates further research into practical challenges, including the development of a dynamic system model and addressing the uncertainties, such as ambient temperature fluctuations, solar radiation variations, and other weather-related factors. These uncertainties can significantly diminish the effectiveness of MPC in real-world scenarios, complicating the optimization process and increasing the overall complexity of the control problem. Addressing these issues is crucial for enhancing the reliability and performance of MPC in HVAC applications [4,8]. The present study therefore proposes a stochastic MPC framework that explicitly accounts for parameter uncertainty, going beyond the reactive nature of PID and the deterministic assumptions of conventional MPC.

1.1.3. Model uncertainty

Model uncertainties arise from the imperfect approximations and simplifications of the actual systems [16]. Ryzhov et al. [17] emphasize that directly applying MPC strategies may not yield significant energy savings due to the inherent variances in building characteristics and contexts, highlighting the necessity for customized solutions.

Hand-tuned controllers often lead to sub-optimal performance and non-adaptivity, failing to adjust to external factors like weather changes. Research into stochastic adaptive control has introduced systematic model adaptation within closed-loop systems, enabling real-time adjustments of MPC parameters to reflect actual performance [18]. Frameworks developed by Maasoumy et al. [16,19] and Xu et al. [20] illustrate the significant effectiveness of adaptive control methods in building energy management. These approaches involve dynamically adjusting controller parameters in response to relevant environmental conditions, which not only enhance energy efficiency but also ensure occupant comfort.

Adaptive control approaches require real-time adjustments to ensure optimal performance. As a result, numerous studies have focused on analyzing MPC parameters using historical data. The parameters obtained from this analysis have been incorporated into various MPC frameworks to effectively mitigate the impacts of uncertainty. These frameworks include sensitivity analysis [21], polytopic MPC [12], and chance-constrained MPC [7,15].

Wang et al. [21] examine how uncertainties in model parameters impact the performance of MPC in building systems. By introducing varying levels of uncertainty in the parameters of thermal resistor-

capacitor model within a virtual test case, the study evaluates MPC's effectiveness in optimizing heat pump signals to reduce energy costs while ensuring occupant comfort. The findings reveal that negative deviations in parameter identification significantly hinder MPC performance, while positive deviations have little effect. The research also offers recommended parameter ranges to maintain MPC effectiveness within a 5% to 10% variability, thereby providing insights into the relationship between model uncertainties and control decisions in building management.

Nagpal et al. [12] introduce a robust controller for building climate control that effectively manages parametric uncertainties in the MPC framework. The controller accounts for uncertainties through polytopic representations and minimizes a worst-case cost function, leading to a min-max optimization problem solved using linear matrix inequalities. Simulation results show that the robust controller successfully maintains desired indoor temperatures despite significant uncertainties, outperforming a nominal controller by 24% under normal conditions and by 17% during severe uncertainty scenarios. Mohebi et al. [15] introduce a chance-constrained stochastic MPC framework to enhance building energy efficiency by addressing uncertainties in thermal control. The framework utilizes historical data, Monte Carlo simulations, and optimization to evaluate energy costs and thermal discomfort. Validation with a residential building case demonstrates that CS-MPC outperforms deterministic MPC by eliminating thermal discomfort, with a potential energy cost increase of up to 7.53%.

These studies demonstrate the importance of handling parameter uncertainty, but most rely on online adaptation or distributional assumptions. In contrast, the current study develops a non-adaptive stochastic MPC that quantifies RC parameter uncertainty offline through scenario generation and reduction, avoiding real-time parameter identification while maintaining computational tractability.

1.1.4. Methods to handle uncertainties

In addressing uncertainties within building energy systems, methods such as scenario-based stochastic programming, robust optimization, and chance-constrained programming are commonly employed. Each of these methodologies offers distinct advantages and challenges, making their appropriateness contingent on the specific context in which they are applied [22].

Scenario-based stochastic programming is used to address uncertainties by creating multiple scenarios based on different probabilistic distributions of uncertain factors. It allows decision-makers to assess outcomes across various potential situations. A key advantage of this approach is its adaptability to different types of uncertainties, such as fluctuations in demand or variability in supply. However, it can be computationally demanding, particularly as the number of scenarios grows, which may strain resources in large-scale situations. Furthermore, the accuracy of the chosen probability distributions is vital; incorrect assumptions can result in poor decision-making [22–24]. Pippia et al. [25] compared the performance of deterministic MPC with a stochastic scenario-based MPC (SBMPC) approach using a nonlinear Modelica model. By incorporating multiple disturbance scenarios, SBMPC aims to improve control robustness and accuracy in building dynamics. The research shows that the SBMPC controller outperforms conventional methods in simulations, highlighting its effectiveness in building energy management for reducing costs and emissions.

Robust optimization, on the other hand, seeks to deliver solutions that perform effectively across a variety of potential scenarios without the need for probability distributions. It emphasizes preparing for the worst-case scenario within a specified range of feasible parameter variations, providing solutions that protect against extreme uncertainties. However, a significant challenge of robust optimization is its tendency to be overly cautious; solutions that are satisfactory in worst-case scenarios may not be optimal in more favorable conditions, potentially resulting in higher operational costs during stable periods [23,24,26]. Yang et al. [27] present an Adaptive Robust MPC (ARMPC) method

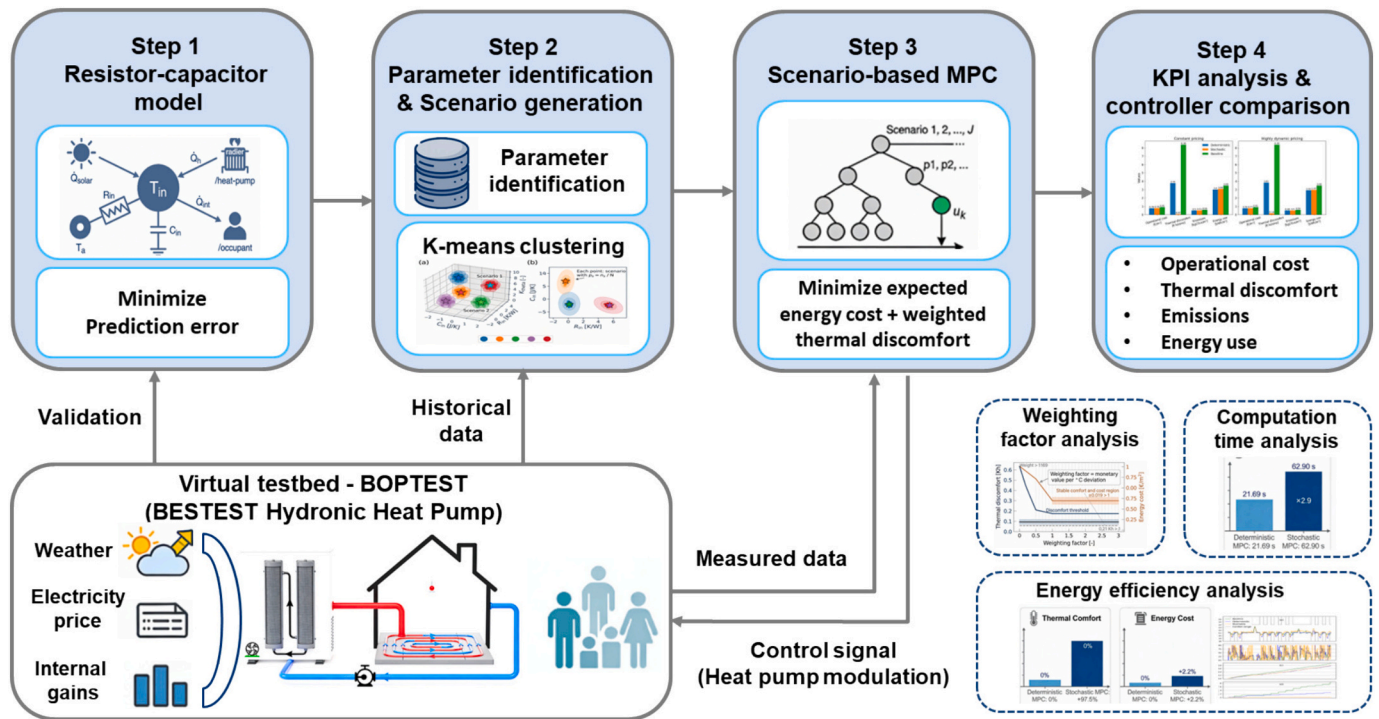


Fig. 1. Overview of the methodology employed in this study, consisting of four steps: (1) development of a resistance–capacitance model for building thermal dynamics; (2) parameter identification using a non-convex model to generate and refine RC parameter scenarios from historical data; (3) implementation of a stochastic MPC model that integrates these scenarios with their probabilities; and (4) performance comparison of the stochastic framework against deterministic MPC and a baseline PI controller, focusing on energy cost and thermal comfort.

designed to improve building models and handle uncertainties in disturbances. By integrating adaptive modeling with robust optimization, ARMPc surpasses traditional MPC controllers and thermostats in maintaining thermal comfort, while also achieving significant energy savings, particularly in scenarios with fluctuating internal load uncertainties.

Chance-constrained programming integrates risk directly into the optimization framework by allowing certain constraints to be violated within a specified probability. This approach enables a nuanced balance between costs and risks associated with uncertainties [22,28]. By permitting controlled risk acceptance, it facilitates more aggressive optimization strategies tailored to user-defined tolerance levels. However, this methodology necessitates modeling of the probability distributions of uncertainties, which can complicate the optimization process, particularly when the underlying distributions are unknown or inaccurately characterized [29]. Mohebi and Wang [7] proposed an optimization model with chance constraints to manage parameter uncertainties and evaluate their impact on MPC performance regarding energy costs and thermal comfort. By adjusting the confidence level, the methodology enables tighter comfort boundaries, influencing thermal discomfort and energy costs effectively.

While these methods either require online adaptation, depend on distributional assumptions, or become computationally heavy with large scenario sets. The present study addresses these drawbacks by performing offline scenario reduction using silhouette-guided k-means clustering, which reduces computation time while preserving the key features of parameter uncertainty.

1.2. Research gaps

During the literature review, two main gaps were identified:

- Despite considerable research on the effects of disturbances on MPC performance, there remains a lack of thorough investigation into

how model uncertainties impact control effectiveness. Most MPC frameworks depend on static model parameters, such as those derived from resistor-capacitor models. Critical parameters, including thermal capacitance, which influences temperature changes; resistance to heat exchange; effective window area; and heat gain from occupants, are essential for accurate modeling. However, these parameters can significantly diverge from their actual values, leading to suboptimal performance and limiting the overall efficacy of the control system.

- The development of adaptable MPC structures capable of real-time parameter adjustments to accurately reflect the actual conditions of a smart building and address uncertainties can be computationally intensive. This complexity requires an additional layer of parameter identification at each time step, complicating the control process and potentially leading to delays in decision-making. As a result, there is an urgent need to establish non-adaptive MPC frameworks that can effectively manage uncertainties while ensuring efficient computation times.

To fill these gaps, this study proposes a non-adaptive stochastic MPC framework that quantifies RC parameter uncertainty through offline scenario generation and reduction, eliminating the need for real-time parameter adaptation while maintaining computational efficiency.

1.3. Scope and objectives

To address the identified gaps, this study quantifies the impact of model uncertainty on MPC performance using a virtual test case. The specific objectives are:

- Obtain RC model parameter scenarios by using historical data, parameter identification methods, and scenario reduction techniques, thereby reducing computational time.

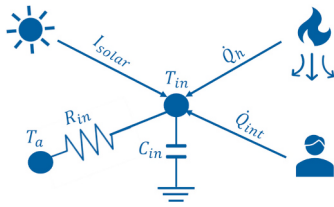


Fig. 2. Schematic representation of a single thermal zone building modeled using a 1R1C network. This model accounts for the influences of solar radiation, ambient temperature, internal heat gains from occupants, and the HVAC system.

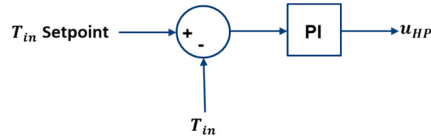


Fig. 3. Illustration of proportional-Integral (PI) controller (BOPTES baseline) [35].

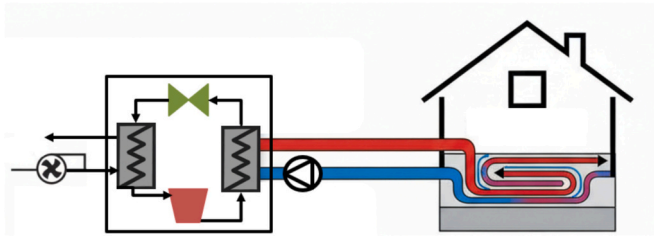


Fig. 4. Illustration of BESTEST Hydronic Heat Pump test case from BOPTES.

- Develop a stochastic MPC framework that integrates multiple RC parameter scenarios and assigns probabilities to each scenario.
- Evaluate and compare the performance of stochastic MPC, deterministic MPC, and a baseline PI controller in terms of thermal comfort and energy cost.
- Analyze the trade-off between energy cost and thermal discomfort and derive implications for designing improved control strategies for building energy management.
- Analyze the trade-off between computation time and the number of reduced scenarios, determining the optimal number that balances computational tractability and preservation of parameter variability.

The main novelty of this study is the integration of non-convex parameter identification, silhouette-guided k-means scenario reduction, and scenario-based stochastic MPC to handle RC parameter

Table 1

Input data of the BESTEST Hydronic Heat Pump test case used in proposed methodology, including building geometry, weather conditions, occupancy, HVAC system specifications, comfort ranges, and the baseline PI controller.

Category	Specification	Value
Building	Location	Belgium
	Floor area	12 × 16 m ²
	Height	2.7 m
	Window area	24 m ²
Weather Condition	Ambient temperature/	Fig. 5(b)
	Solar radiation	
Occupancy	Occupants	5
	Weekdays/Weekends	(20:00–07:00) / (00:00–24:00)
HVAC	Heating source	Air-to-water heat pump (Thermal capacity: 15 kW) (Power capacity: 3.5 kW)
	End Equipment	Floor heating system
	Comfort range	21 – 24° C (Occupied) / 15 – 30° C (Unoccupied)
Baseline	PI controller	Fig. 3

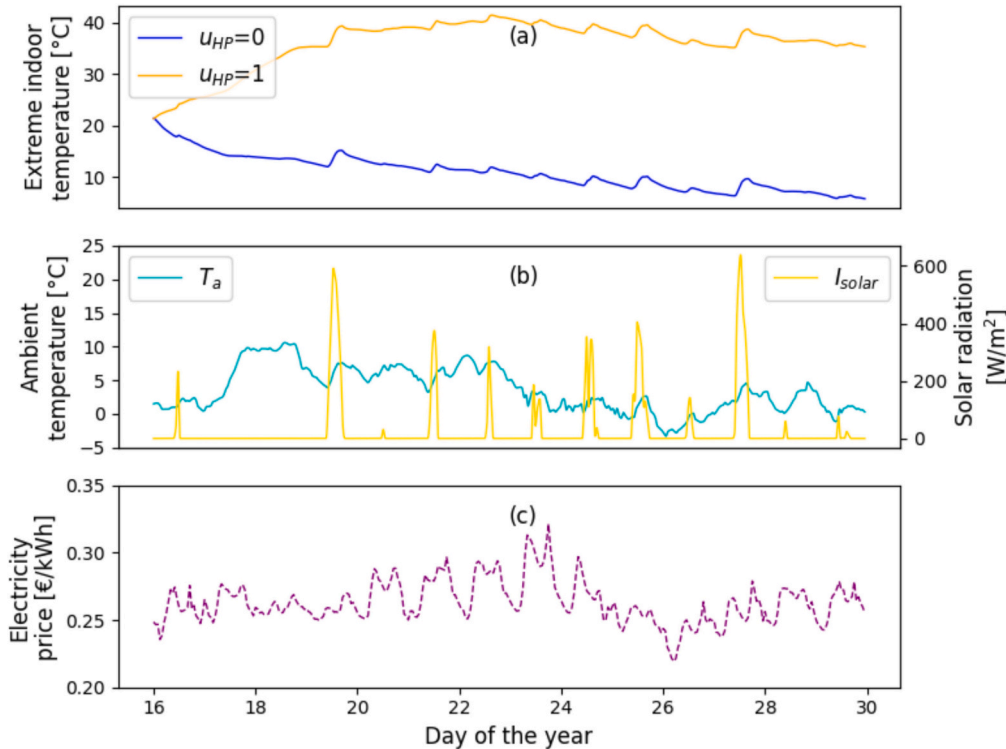


Fig. 5. Hourly data from BOPTES a) the extreme indoor temperatures observed when the heat pump modulation signal (u_{HP}) is set to 0 and 1 during the sampling period. (b) the corresponding weather conditions, including solar radiation and ambient temperature (c) the electricity price evolution.

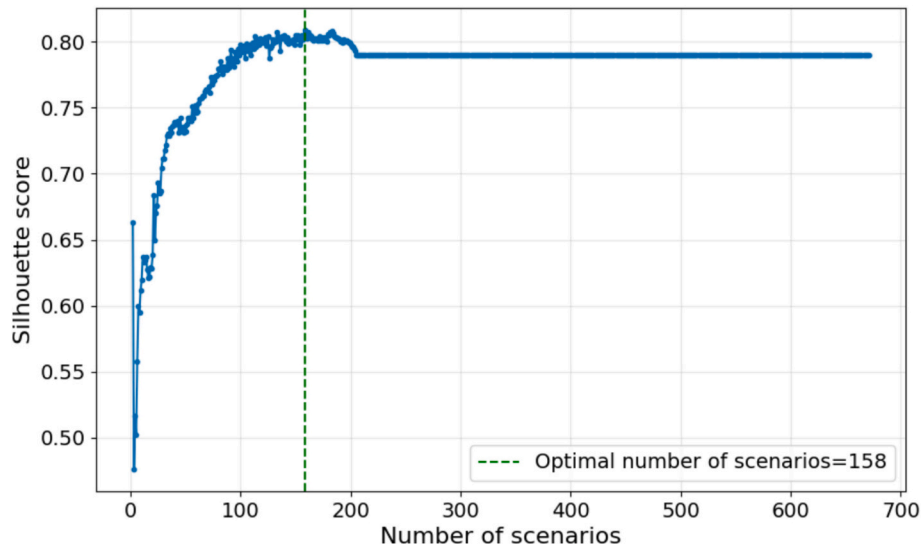


Fig. 6. Silhouette coefficient values for k-means clustering. The maximum average silhouette score occurs at 158 scenarios, indicating that 158 clusters best balance intra-cluster cohesion and inter-cluster separation for the 672 RC parameter scenarios.

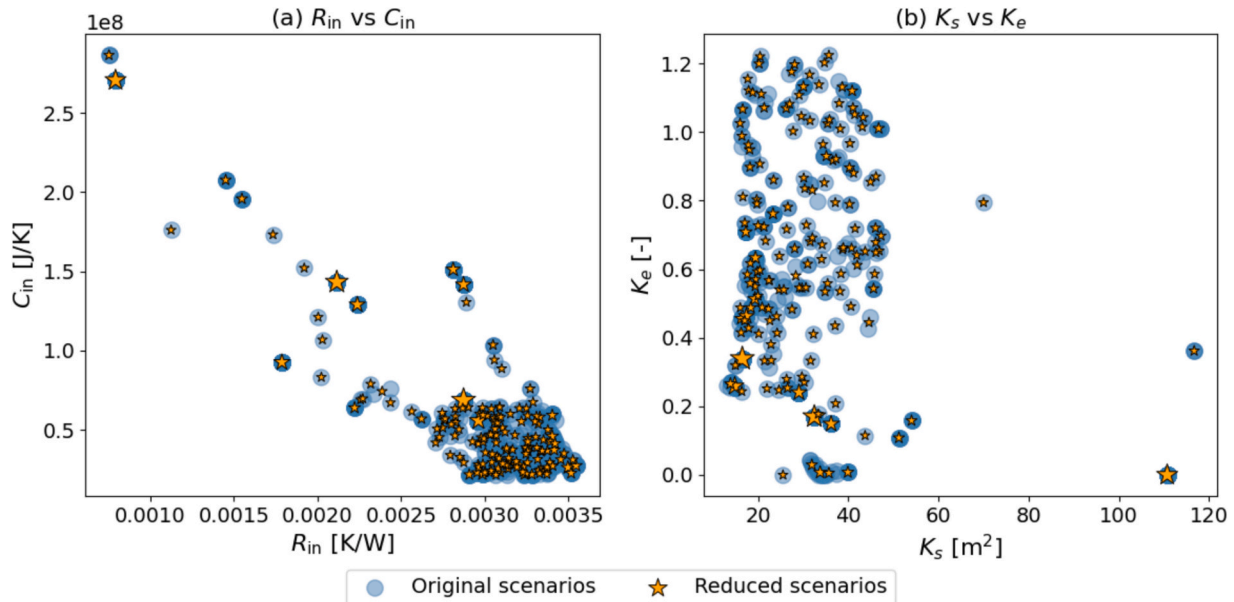


Fig. 7. Scatter plots of RC parameter scenarios: (a) thermal resistance R_{in} versus thermal capacitance C_{in} ; (b) solar gain coefficient K_s versus internal gain coefficient K_e . Circles represent the original 672 parameter sets obtained from hourly identification; stars represent the 158 reduced cluster centroids after k-means clustering, with star sizes proportional to the probability of each reduced scenario (maximum probability = 11.61%).

Table 2

Sample of reduced RC parameter scenarios and their probabilities after K-means clustering.

Scenario	$R_{in} [\frac{K}{W}]$	$C_{in} [\frac{J}{K}]$	$K_s [m^2]$	$K_e [-]$	Probability
1	2.87×10^{-3}	6.86×10^7	1.65×10^1	3.38×10^{-1}	11.61%
	2.11×10^{-3}	1.43×10^8	1.70×10^1	1.70×10^{-1}	
2	10^{-3}	10^8	3.24×10^1	10^{-1}	10.26%
	7.93×10^{-4}	2.71×10^8	1.10×10^1	3.14×10^{-1}	
3	10^{-4}	10^8	10^2	10^{-12}	8.33%
	2.96×10^{-3}	5.58×10^7	1.48×10^1	2.58×10^{-1}	
4	10^{-3}	10^7	10^1	10^{-1}	5.80%
	2.23×10^{-3}	1.29×10^8	2.90×10^1	2.40×10^{-1}	
5	10^{-3}	10^8	10^1	10^{-1}	4.02%
...	...	...	...	...	...

uncertainty offline, enabling tractable real-time control. The primary research question addressed is: Does stochastic MPC with offline reduced scenarios outperform deterministic MPC in terms of thermal comfort and energy cost while preserving computation efficiency?

2. Method

The methodology employed in this study comprises four sequential steps, as illustrated in Fig. 1. First, a first-order resistance-capacitance (1R1C) model is developed to represent the building's thermal dynamics (Section 2.1). Second, parameter identification is performed using an optimization framework applied to 672 h of BOPTEST data, generating a set of 672 RC parameter scenarios that capture epistemic uncertainty arising from weather variations, occupancy patterns, and unmodeled dynamics (Section 2.2). Third, three controllers are implemented: a

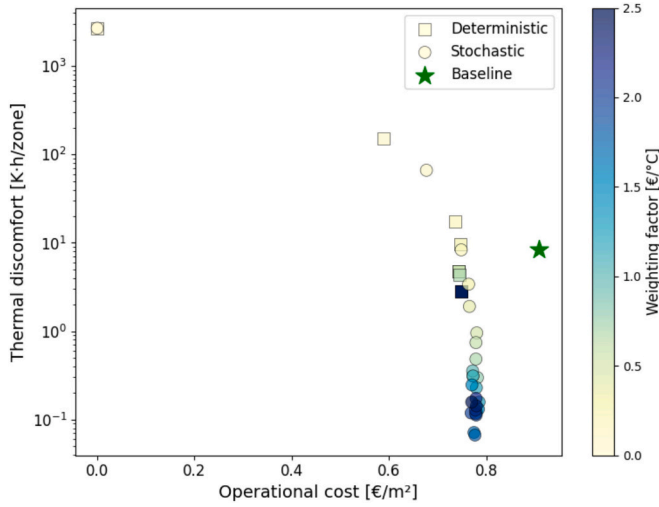


Fig. 8. Performance comparison of the stochastic MPC, deterministic MPC, and baseline PI controller across a range of weighting factors (0–2.5 $\frac{\text{€}}{^\circ\text{C}}$) that balance energy cost and thermal discomfort.

stochastic MPC that uses the reduced scenarios, a deterministic MPC that employs mean parameters as a benchmark, and a PI controller as a reactive baseline (Section 2.3). Fourth, all controllers are evaluated over a two-week validation period using total energy cost and thermal discomfort metrics, systematically quantifying uncertainty effects on building thermal management performance (Section 2.4).

2.1. Building thermal model

To model the thermal behavior of the building, an RC network model is utilized, as depicted in Fig. 2. In this framework, the building is conceptualized as a network where nodes represent walls or individual rooms. Each thermal zone within the building is characterized using a lumped resistance and capacitance approach, which accounts for various components such as the internal thermal mass, indoor air, building envelope, outdoor environment, and the dynamics of heat exchange [4]. The RC model used in this work is expressed in Eq. (1):

$$C_{in} \frac{dT_{in}}{dt} = \frac{T_a - T_{in}}{R_{in}} + K_e \cdot \dot{Q}_{int} + \dot{Q}_h + K_s \cdot I_{solar} \quad (1)$$

In this equation, C_{in} denotes the thermal capacitance of the zone and its envelopes ($\frac{J}{K}$), while R_{in} represents the resistance to heat exchange between the outdoor environment and the building envelopes ($\frac{K}{W}$). The variables T_a and T_{in} correspond to the outdoor and indoor temperatures (K), respectively. The term I_{solar} signifies the solar irradiation ($\frac{W}{m^2}$), while $\dot{Q}_h$ indicates the heat influx from the heating system (W). Additionally, $\dot{Q}_{int}$ captures the internal heat load generated by occupants, lighting, and indoor equipment (W). The parameter K_s refers to the effective area of the windows (m^2), and K_e denotes the coefficient representing the heat gain from occupants.

2.2. Uncertainty characterization

This step captures and reduces the uncertainty in the RC model parameters. Parameter identification is performed for each hour of the BOPTEST dataset, yielding 672 scenarios that reflect epistemic uncertainty. These scenarios are then reduced to a computationally tractable set using k-means clustering, with the optimal number of clusters determined by silhouette analysis. The resulting centroids and their associated probabilities serve as inputs to the stochastic MPC.

2.2.1. Parameter identification

This paper utilizes a parameter identification method grounded in the non-convex optimization model presented in [1]. By obtaining scenarios for the RC parameters, more realistic values can be derived that effectively reflect the impact of external factors, such as solar radiation and ambient temperature, as well as internal conditions, including heat flux from the heating system and heat gains from occupants, on the thermal dynamics of buildings. To generate initial parameter values, Latin hypercube sampling [30] is used with 1000 samples per identification run and selecting the solution with the lowest least-squares error, ensuring a comprehensive exploration of the parameter space and avoiding poor local optima. The objective function is specifically designed to minimize the least squares of the discrepancies between the state variables and the corresponding measured data, ensuring that our model aligns closely with actual building performance. The parameter identification model is detailed in Eqs. (2)–(5):

$$\min_{R_{in}, C_{in}, K_s, K_e} \sum_{t=1}^N \left(T_{in}^t - \tilde{T}_{in}^t \right)^2 \quad (2)$$

$$\tilde{T}_{in}^1 = \left(1 - \frac{h}{R_{in} C_{in}} \right) \cdot T_{in}^0 + \left[\frac{h}{R_{in} C_{in}} \quad \frac{h}{C_{in}} \quad \frac{h K_s}{C_{in}} \quad \frac{h K_e}{C_{in}} \right] \cdot [T_a^0 \quad Q_h^0 \quad I_{solar}^0 \quad Q_{int}^0]^T \quad (3)$$

$$\tilde{T}_{in}^{t+1} = \left(1 - \frac{h}{R_{in} C_{in}} \right) \cdot \tilde{T}_{in}^t + \left[\frac{h}{R_{in} C_{in}} \quad \frac{h}{C_{in}} \quad \frac{h K_s}{C_{in}} \quad \frac{h K_e}{C_{in}} \right] \cdot [T_a^t \quad Q_h^t \quad I_{solar}^t \quad Q_{int}^t]^T, t=1, \dots, N \quad (4)$$

$$R_{in}, C_{in}, K_s, K_e \geq 0 \quad (5)$$

Here, h represents the step size, T_{in}^t denotes the measured indoor temperature, and $\tilde{T}_{in}^t$ indicates the modeled indoor temperature at each time step t . R_{in} and C_{in} are correlated because the time constant $h = R_{in} C_{in}$ may dominate the system response. The ensemble of 672 identified sets captures the spread of this correlation, which is later preserved in the scenario reduction step.

2.2.2. Scenario generation

The parameter identification process described in section 2.2.1, is applied to each hour of the 672-h BOPTEST data, which covers one month period. This yields a set of 672 scenarios, each representing a plausible realization of the RC model parameters $\theta_t = [R_{in,t}, C_{in,t}, K_{s,t}, K_{e,t}]$ governing building thermal dynamics under uncertainty. While this ensemble captures the epistemic uncertainty arising from unmodeled thermal dynamics and measurement variability, its size is prohibitively large for real-time stochastic MPC.

2.2.3. Scenario reduction

To obtain a computationally tractable representation while preserving the diversity of the original parameter distribution, a scenario reduction framework based on clustering is employed. K-means clustering is used to partition the full set of scenarios into a specified number of groups, with the optimal number of clusters determined via silhouette analysis.

2.2.3.1. K-means clustering. The k-means algorithm partitions the standardized parameter matrix into k clusters by minimizing total within-cluster sum of squared Euclidean distances denoted in Eq. (6) and Eq. (7). Each cluster C_k is represented by its centroid μ_k , defined as the mean of the points assigned to that cluster [31]. The objective function is

$$W(C_k) = \sum_{x_i \in C_k} (x_i - \mu_k)^2 \quad (6)$$

where x_i is i^{th} data point of cluster k (C_k), and the total within-cluster

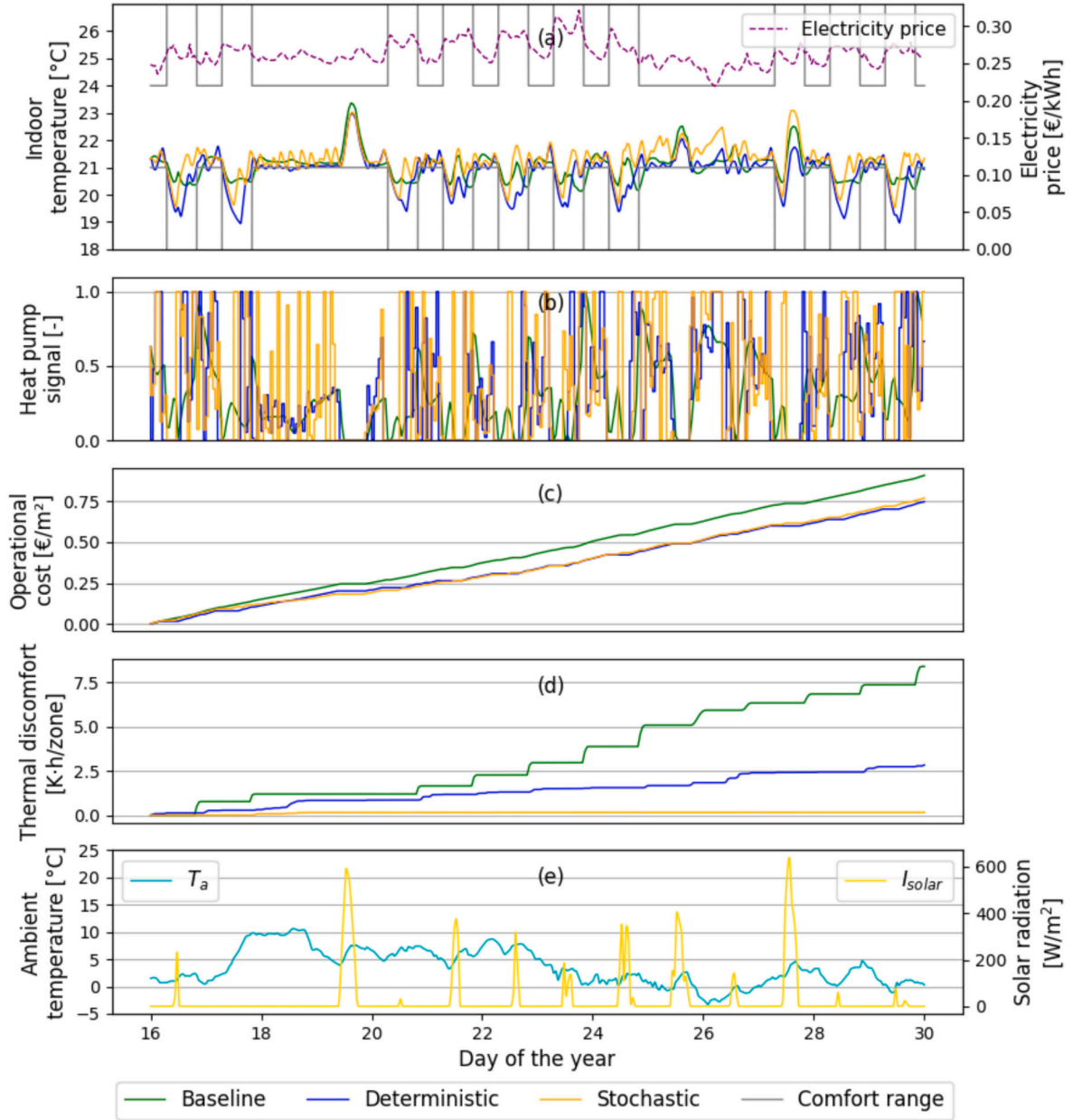


Fig. 9. Simulation results for all control strategies at a weighting factor setting of $2.5 \frac{\text{€}}{\text{C}}$: a) indoor temperature and comfort bounds; b) heat pump modulation signal u_{hp} ; c) cumulative energy cost; d) cumulative thermal discomfort; e) weather conditions (ambient temperature and solar radiation). The stochastic MPC maintains comfort more effectively and achieves the lowest cumulative cost and discomfort.

variation is

$$W(C) = \sum_{k=1}^k W(C_k) = \sum_{k=1}^k \sum_{x_i \in C_k} (x_i - \mu_k)^2 \quad (7)$$

In summary, the K-means clustering process involves following key steps: first, the number of clusters k is predetermined; next, k initial centroids are randomly selected from the dataset. Subsequently, each data point is assigned to the closest centroid, and the new centroids are computed based on the average of the assigned data points. This iterative process continues until the centroids stabilize and no further changes occur.

2.2.3.2. Optimal number of scenarios. The optimal number of clusters k^* , is determined using silhouette analysis [32], an internal validation technique that quantifies cluster quality shown in Eq. (8) and Eq. (9). For

a candidate k , the parameter matrix is standardized to ensure equal weighting of parameters. After applying k-means, the silhouette coefficient $s(i)$ for each data point i is computed as

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (8)$$

where $a(i)$ is the mean Euclidean distance from sample i to all other points in its cluster, and $b(i)$ is the minimum mean distance to samples in any other cluster. The average silhouette coefficient is then

$$\bar{S}(k) = \frac{1}{N_{\max}} \sum_{i=1}^{N_{\max}} s(i) \quad (9)$$

$N_{\max}$ is the total number of original scenarios. $\bar{S}(k)$ ranges from -1 to 1 , with higher values indicating better-defined clusters. The optimal number of clusters is selected as $k^* = \arg \max \bar{S}(k)$. For each cluster, the

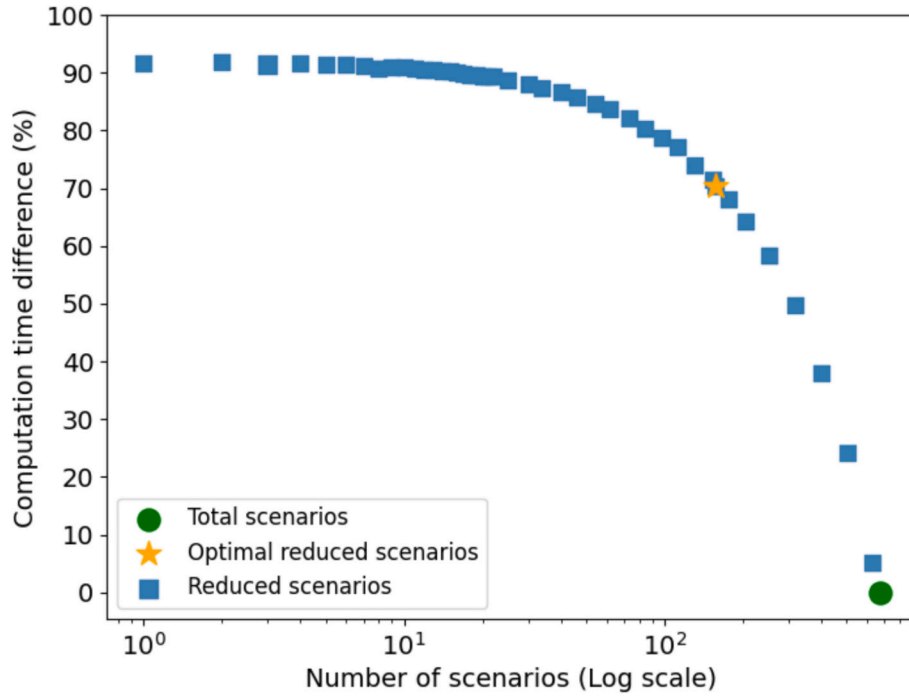


Fig. 10. Computational time reduction of scenario-reduced stochastic optimization relative to the full set of 672 scenarios, plotted against the number of reduced scenarios

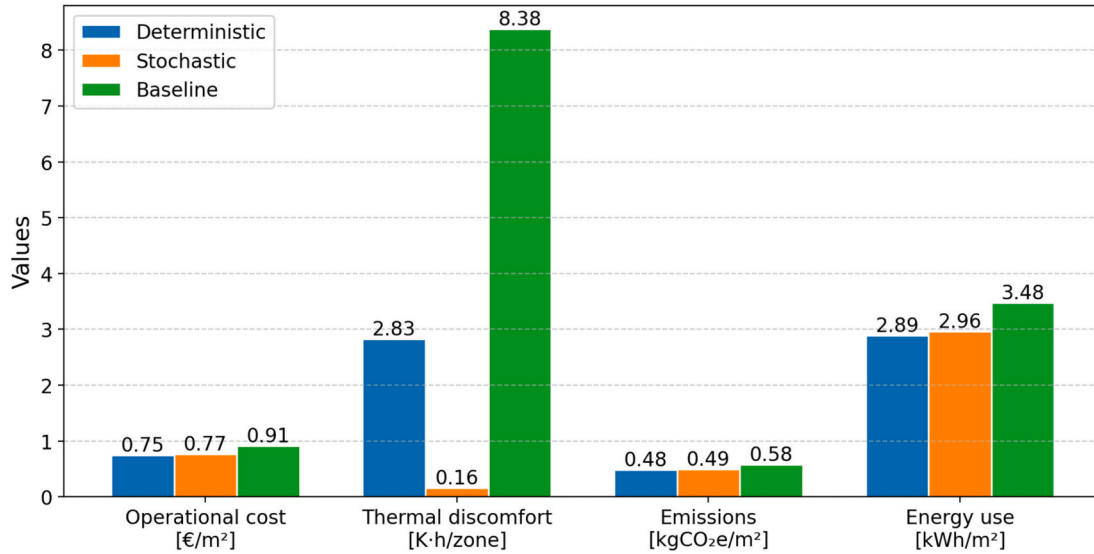


Fig. 11. Comparison of KPIs for deterministic MPC, stochastic MPC, and the baseline PI controller, including emissions, energy use, operational cost, and thermal discomfort.

probability of its corresponding reduced scenario is calculated based on the number of original scenarios as shown in Eq. (10):

$$Prob^k = \frac{N_k}{N_{max}} \quad (10)$$

where N_k is the number of original scenarios in cluster k .

2.3. Controller Formulation

Three controllers are implemented to evaluate the benefits of accounting for parameter uncertainty: a deterministic MPC using average parameters (benchmark), a stochastic MPC using the reduced scenarios,

and a baseline PI controller.

2.3.1. Deterministic MPC

The deterministic MPC controller serves as a benchmark in this study. It employs a single set of RC parameters derived by averaging the 672 identified parameter sets obtained from the scenario generation step, thereby representing the nominal building dynamics. This averaged model provides a basis for evaluating the added value of the stochastic MPC approach, which explicitly accounts for parameter uncertainty.

The controller minimizes energy costs while addressing thermal discomfort, as outlined in Eq. (11). The system is modeled using discrete-time linear state-space equations with a defined prediction

Table 3

Comparative analysis of control strategies for the BOPTEST BESTEST Hydronic Heat Pump test case, reporting operational cost, thermal discomfort, total weighted cost, and improvement over the PI baseline.

Study	Methodology	Operational cost ($\frac{\text{€}}{\text{m}^2}$)	Thermal discomfort ($\frac{\text{Kh}}{\text{zone}}$)	Total weighted cost (€)	Improvement compared to PI baseline (%)
Wang et al. [36]	Reinforcement learning (Deep deterministic policy gradient)	0.81	0.75	157.02	17.99%
Mohebi et al. [15]	Chance-constraint MPC (Different uncertainty levels) (Linear error prediction)	[0.75–0.8]	[0–5.80]	[148.14–154.65]	19.23–22.63%
Mohebi et al. [15]	Chance-constraint MPC (Different uncertainty levels) (Non-linear error prediction)	[0.75–0.8]	[0–5.63]	[147.24–154.50]	19.31–23.10%
BOPTEST baseline [33]	PI controller	0.91	8.38	191.48	–
Current study	Scenario-based stochastic MPC	0.77	0.16	148.17	24.18%

horizon, (H). To achieve this objective, several key constraints must be satisfied. The indoor temperature for the next time step is calculated through Eq. (12), which reflects the building's thermal dynamics. Eq. (13) ensures that the indoor temperature remains within specified limits to uphold occupant comfort. The initial indoor temperature is established based on the ground truth value, as indicated in Eq. (14). Furthermore, the power consumption and heat production of the HVAC system are defined by Eq. (15) and Eq. (16), respectively. The control input is constrained within the range of 0 to 1, as specified in Eq. (17), while the slack variable, accounting for deviations from the desired temperature range, must remain non-negative, as outlined in Eq. (18).

$$\min_{u_{hp}} \sum_{t=n}^{n+H} (P_e^t \cdot P_{HP}^t + w \cdot s^t) \quad (11)$$

s.t

$$\tilde{T}_{in}^{t+1} = \left(1 - \frac{h}{R_{in}C_{in}}\right) \cdot \tilde{T}_{in}^t + \left[\frac{h}{R_{in}C_{in}} \frac{h}{C_{in}} \frac{hK_s}{C_{in}} \frac{hK_e}{C_{in}}\right] \cdot [T_a^t \ Q_{HP}^t \ I_{solar}^t \ Q_{inter}^t]^T \quad (12)$$

$$T_{min}^t - s^t \leq \tilde{T}_{in}^t \leq T_{max}^t + s^t \quad (13)$$

$$\tilde{T}_{in}^0 = T_g \quad (14)$$

$$P_{hp}^t = 1290 \cdot u_{HP}^t + 1115 \quad (15)$$

$$Q_{hp}^t = 7221 \cdot u_{HP}^t + 2220 \quad (16)$$

$$0 \leq u_{HP}^t \leq 1 \quad (17)$$

$$s^t \geq 0 \quad (18)$$

In this framework, P_e denotes the energy price, while P_{HP} represents the power consumption of the heat pump at each time step t . The weighting factor w balances the trade-off between energy costs and thermal discomfort, and s^t serves as a slack variable for deviations from the acceptable temperature range. Q_{HP}^t indicates the heat production of the heat pump, and T_g is the ground truth temperature at the simulation's start. The control output of the MPC, specifically the modulation signal for the heat pump, is represented by u_{HP}^t . T_{min}^t and T_{max}^t are model inputs, defining the essential lower and upper limits of the zone temperature for occupant comfort. The temperature bounds follow the BOPTEST framework [33], which adopts the thermal comfort criteria of ASHRAE Standard 55 [34].

2.3.2. Stochastic MPC

In this study, the stochastic MPC formulation (Eq.(19)–(27)), extends the deterministic framework by explicitly considering J distinct scenarios, each representing a plausible realization of the RC model

parameters $\theta = [R_{in}, C_{in}, K_s, K_e]$ obtained from the scenario reduction step (Section 2.4). Each scenario j is assigned a probability $Prob^j$ proportional to the number of original scenarios it represents, reflecting its likelihood in the identified uncertainty distribution. As shown in Eq. (19), the objective is to minimize the expected cost over all scenarios.

$$\min_{u_{hp}} \sum_{k=n}^{n+H} \sum_{j=1}^J Prob^j \cdot (P_e^t \cdot P_{HP}^{t,j} + w \cdot s^{t,j}) \quad (19)$$

s.t

$$\tilde{T}_{in}^{t+1,j} = \left(1 - \frac{h}{R_{in}^j C_{in}^j}\right) \cdot \tilde{T}_{in}^{t,j} + \left[\frac{h}{R_{in}^j C_{in}^j} \frac{h}{C_{in}^j} \frac{hK_s^j}{C_{in}^j} \frac{hK_e^j}{C_{in}^j}\right] \cdot [T_a^t \ Q_{HP}^t \ I_{solar}^t \ Q_{inter}^t]^T \quad (20)$$

$$T_{min}^t - s^{t,j} \leq \tilde{T}_{in}^{t,j} \leq T_{max}^t + s^{t,j} \quad (21)$$

$$\tilde{T}_{in}^{0,j} = T_g \quad (22)$$

$$P_{hp}^{t,j} = 1290 \cdot u_{HP}^{t,j} + 1115 \quad (23)$$

$$Q_{hp}^{t,j} = 7221 \cdot u_{HP}^{t,j} + 2220 \quad (24)$$

$$0 \leq u_{HP}^{t,j} \leq 1 \quad (25)$$

$$u_{HP}^{t,j} = \beta^t \quad (26)$$

$$s^{t,j} \geq 0 \quad (27)$$

It is a common approach to calculate the expectation of the optimal strategy set derived from various scenarios and use this as the control action. This is mathematically represented as $u_{HP}^t = \frac{\sum_{k=1}^K Prob^k \cdot u_{HP}^{t,k}}{K}$. However, relying on the averaged strategy can lead to conflicts among the scenario-dependent optimal strategies, potentially resulting in an infeasible control problem [30]. To address this, we implemented a revised approach that introduces a key constraint into the MPC framework, as outlined in Eq. (26). This constraint forces all scenarios to use the same control input. Here, β^k is a shared decision variable representing the final optimal control strategy.

2.3.3. Baseline PI controller

In this research, an embedded PI controller is employed as a baseline to evaluate and compare developed control algorithms. The tuning parameters are documented in the BOPTEST repository [35]. This controller considers the indoor temperature, (T_{in}), as its input and generates the heat pump modulation signal, (u_{hp}), as its output, as illustrated in Fig. 3. The primary objective of the PI controller is to regulate the indoor temperature, steering it toward the designated

setpoint.

2.4. Performance evaluation

All controllers—deterministic MPC, stochastic MPC, and the baseline PI controller—are evaluated on the BESTEST Hydronic Heat Pump test case provided by the Building Optimization Testing Framework (BOPTEST) [35].

2.4.1. Evaluation metrics

The evaluation is performed over a two-week validation period, which corresponds to the “peak heating” scenario defined in BOPTEST (January 17–31). This period is chosen because it features high heating demand and thus clearly exposes differences in the controller's performance. The controllers are compared using two core Key Performance Indicators (KPIs) defined in BOPTEST including energy cost and thermal discomfort. KPIs represented in Eq. (28) and Eq. (29) are calculated using the measurement data stored during the simulation.

2.4.1.1. Thermal discomfort. The thermal discomfort is quantified as the cumulative deviation of the indoor temperature from the predefined comfort bounds, measured in kelvin-hours ($K.h$). The slack variable s^t in the objective function of MPC in Eq. (11), directly penalizes this deviation. The reported thermal discomfort after simulation is computed as

$$Discomfort = \frac{\sum_{z=1}^N \int_{t_s}^{t_f} \left(\max(\tilde{T}_{in}^t - T_{max}^t, 0) + \max(T_{min}^t - \tilde{T}_{in}^t, 0) \right) dt}{N} \quad (28)$$

where t_s and t_f are the start and end times of the evaluation period, and N is the number of zones (here $N = 1$) [33].

2.4.1.2. Energy cost. This metric quantifies the operational expense normalized by building area. It is calculated as

$$Cost = \frac{\int_{t_s}^{t_f} P_e^t \cdot P_{total}^t \cdot dt}{A} \quad (29)$$

where P_e^t is the electricity price, P_{total}^t is the total electrical power consumption of the heat pump, circulation pump, and fan, and A is the floor area.

2.4.2. Target system

The test case represents a single-zone residential structure located in Belgium, as illustrated in Fig. 4. This building features a floor area of $12 \times 16 \text{ m}^2$ and a height of 2.7 m, with a window area of 24 m^2 that enhances natural light and ventilation. Designed to accommodate five occupants, the building has specific occupancy patterns, with residents present on weekdays from 20:00 to 07:00 and throughout the day on weekends. The HVAC system is powered by an efficient air-to-water heat pump, boasting a thermal capacity of 15 kW and a power capacity of 3.5 kW, which delivers heating through an integrated floor heating system. To ensure optimal comfort, the indoor temperature is maintained within a range of 21–24 °C when occupied and 15–30 °C when unoccupied.

Fig. 5 shows the hourly ambient temperature and solar radiation, together with the resulting indoor temperature extremes for continuously operating versus switched-off heat pump operation, and the corresponding electricity price scenarios [1,35].

The input data used in the proposed approach is summarized in Table 1. This table outlines the essential properties of the BESTEST Hydronic Heat Pump test case, providing a comprehensive overview of the building specifications, weather conditions, occupancy patterns, HVAC system details, and baseline control strategy.

3. Results

The results are organized into two sections. Section 3.1 presents the RC parameter scenarios obtained from parameter identification, including the generated scenarios for stochastic optimization, both before and after scenario reduction. Section 3.2 presents the performance results of the stochastic MPC and compares them with conventional approaches.

3.1. RC parameters

This study investigates the distributions of the RC parameters by leveraging historical data from the BOPTEST test case and employing the non-convex parameter identification method described in Section 2.2. The identification is performed with a 24-h optimization horizon, yielding 672 parameter sets that capture the epistemic uncertainty.

To facilitate scenario reduction, K-means clustering is implemented to group similar data points based on the four features R_{in} , C_{in} , K_s , and K_e . The optimal number of clusters is determined using silhouette analysis, which evaluates cluster cohesion and separation. As shown in Fig. 6, the silhouette coefficient reaches a maximum at $k^* = 158$, indicating that 158 clusters best represent the underlying structure of the parameter space while maintaining computational tractability. This approach yields a reduced dataset that accurately captures the essential characteristics of the original data while significantly reducing its dimensionality.

The original and reduced scenarios are presented in Fig. 7. Each reduced scenario is assigned a probability proportional to the size of its corresponding cluster, reflecting its prevalence in the original distribution. A sample of the reduced scenarios and their probabilities is given in Table 2. These reduced scenarios and their associated probabilities serve as the input to the stochastic MPC formulation.

3.2. Controller performance comparison

The performance of the proposed stochastic MPC in terms of thermal discomfort and energy cost is illustrated in Fig. 8. The results are compared against both the deterministic MPC and the BOPTEST baseline (PI controller) across a range of weighting factors w from 0 to $2.5 \frac{\text{€}}{\text{°C}}$, which balances the trade-off between energy cost and comfort. Implementing MPC in either deterministic or stochastic form substantially improves energy performance and thermal comfort relative to the PI baseline.

When the weighting factor $w = 0$, the controller disregards thermal comfort entirely, resulting in the lowest energy cost but the highest thermal discomfort. As w increases, energy cost rises while thermal comfort improves, reflecting the inherent trade-off between the two objectives. The stochastic MPC achieves superior thermal comfort relative to the PI baseline at a lower weighting factor ($w > 0.2 \frac{\text{€}}{\text{°C}}$) compared to the deterministic MPC, which requires a higher weighting factor ($w > 0.4 \frac{\text{€}}{\text{°C}}$). This indicates that the explicit handling of parameter uncertainty enables the stochastic MPC to more effectively balance energy cost and comfort, delivering comfort improvements even when the penalty on discomfort is relatively small. Consequently, the stochastic MPC provides a more robust and efficient control solution across a wider range of operational preferences.

Fig. 9 presents the simulation results for all control strategies, including stochastic MPC, deterministic MPC, and the baseline, under the dynamic pricing scenario. The results illustrate the evolution of indoor temperature, energy costs, and thermal discomfort throughout the sampling period. The stochastic MPC maintains the indoor temperature within the comfort band more consistently than the other controllers, and its cumulative discomfort rise at the slowest rate.

4. Discussion

This section begins by analyzing the effect of the weighting factor on the trade-off between energy costs and thermal discomfort. Following this, a comparative analysis of the stochastic MPC framework against previous methodologies is presented, highlighting its competitive performance in terms of total weighted cost including operational costs and thermal discomfort. Finally, the limitations and future work of current study are acknowledged.

4.1. Scenario reduction trade-off

The scenario reduction step (Section 2.2.3) transforms the full set of 672 identified parameter sets into a smaller, computationally tractable set of cluster centroids. Fig. 10 illustrates the trade-off between the number of reduced scenarios and the corresponding reduction in computation time compared to solving the stochastic MPC problem with all 672 original scenarios.

As illustrated in Fig. 10, a small number of reduced scenarios leads to substantial computational savings—for instance, employing only 10 scenarios decreases computation time by approximately 90.8% compared to the full set. However, such an aggressive reduction may neglect significant parameter variability. As the number of scenarios increases, the relative time savings decline, since additional scenarios introduce more optimization variables and constraints. The optimal number of clusters, identified through silhouette analysis, yields a computation time reduction of about 70.4%. This setting offers an effective trade-off: it significantly alleviates computational load relative to the 672-scenario benchmark while maintaining the essential statistical characteristics of the parameter distribution, as confirmed by the controller performance evaluation in Section 3.2.

The trade-off between computation time and thermal comfort is evident: the stochastic MPC (158 scenarios) takes 77.71 s to compute but reduces discomfort to $0.16 \frac{Kh}{zone}$, whereas the deterministic MPC computes in only 21.62 s but results in a discomfort of $2.83 \frac{Kh}{zone}$. These times were measured using CPLEX on an Intel Core i7 (2.1 GHz, 16 GB RAM).

4.2. Comparative analysis

Fig. 11 indicates that, relative to deterministic MPC, the stochastic MPC achieves the lowest thermal discomfort ($0.16 \frac{Kh}{zone}$), representing a 94.42% reduction compared to deterministic MPC. However, energy use and associated emissions increase marginally (by 2.9%) relative to deterministic MPC, reflecting the trade-off between comfort and energy efficiency. Table 3 compares previous works employing different methodologies to evaluate operation costs, thermal discomfort, total weighted costs, and performance improvements. Wang et al. [33] applied reinforcement learning techniques, specifically comparing Deep Deterministic Policy Gradient (DDPG), Dueling Deep Q Networks (DDQN), and Soft Actor Critic (SAC), with DDPG demonstrating the best performance among the three. This approach resulted in an operation cost of $0.81 \frac{€}{m^2}$, the thermal discomfort at $0.75 \frac{Kh}{zone}$, and the total weighted cost of 157.02 € and 17.99% overall improvement compared to the BOPTEST baseline. Mohebi et al. [15] utilized chance-constraint MPC with both linear and non-linear error predictions considering different confidence levels, achieving operation costs ranging from 0.75 to $0.8 \frac{€}{m^2}$, thermal discomfort levels between 0 and 5.80, and total weighted costs between 148.14 and 154.65 €, with improvements of 19.23% to 23.10%. In contrast, the current study employed scenario-based stochastic MPC, demonstrating a favorable operation cost of $0.77 \frac{€}{m^2}$, minimal thermal discomfort at $0.16 \frac{Kh}{zone}$, a competitive total weighted cost of 148.17 €, and a significant improvement of 24.18%. Overall, the current study stands out for its balanced approach, combining low costs and discomfort, making it a compelling option

among the methodologies analyzed.

4.3. Limitation and future work

Based on the available data from BOPTEST, in this study, 1R1C model is selected to simulate the thermal behavior of the building in the BESTEST Hydronic Heat Pump test case. This choice was made because higher-order RC models would introduce additional complexities, such as wall temperature variations, for which we currently lack sufficient data. Moreover, although the findings are promising, they are derived from a single residential building in a specific climate. Further validation across different building types, climates, and seasons and longer-term periods is therefore necessary to assess the generalizability of the approach. Since BOPTEST provides precise disturbance forecasts, the current study focuses on addressing uncertainties in the RC model parameters. Future work will extend the framework to also account for uncertainties in disturbances, such as weather and occupancy predictions.

5. Conclusion

Model predictive control (MPC) is an advanced optimization-based control strategy designed to minimize energy consumption while ensuring thermal comfort in buildings. Typically, MPC employs grey-box resistor-capacitor models, where the parameters capturing the effects of internal and external disturbances on the building's thermal behavior are critical. This paper presents a stochastic MPC framework that includes parameter scenarios derived from historical data to account for uncertainty, enhanced by silhouette-guided k-means clustering to reduce computational burden.

The performance of the proposed stochastic MPC is evaluated against deterministic MPC and a baseline proportional–integral (PI) controller using a single test case within the BOPTEST framework (air-to-water heat pump with floor heating). In this study, results indicate that stochastic MPC achieves a 50% lower discomfort penalty than deterministic MPC, reduces total cost by 24.2% compared with the PI controller over a two-week period, and cuts computation time by 70.4% via clustering. These findings suggest that, for the tested building system and conditions, stochastic MPC can offer meaningful improvements in thermal comfort, energy cost efficiency, and computational feasibility.

CRedit authorship contribution statement

Parastoo Mohebi: Conceptualization, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft. **Ramin Roshandel:** Conceptualization, Investigation, Methodology, Project administration, Supervision, Validation, Writing – review & editing.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work the authors used *Deepseek* in order to polish the language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the of the published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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